Problem 4.36: G-Parity of Mesons

Griffiths, *Introduction to Elementary Particles*

A Detailed, Pedagogical Solution

Problem Statement (Griffiths 4.36)

(a) Using the information in Table 4.6, determine the G-parity of the following mesons:

$$\pi, \quad \rho, \quad \omega, \quad \eta, \quad \eta', \quad \phi, \quad f_2$$

(b) Show that $R_2 \left| I\ 0 \right\rangle = (-1)^I \left| I\ 0 \right\rangle$, and use this result to justify Equation 4.60:

$$G = (-1)^I C \qquad \text{(Equation 4.60)}$$

Table 4.6 data (quantum numbers of light-quark mesons):

ℓ	s	J^{PC}	$I = 1$	$I = \frac{1}{2}$	$I = 0$
0	0	0^{-+}	π	K	$\eta,\ \eta'$
0	1	1^{--}	ρ	K^*	$\omega,\ \phi$
1	0	1^{+-}	B	Q_2	$H,\ ?$
3*1	3*1	0^{++}	δ	κ	$\epsilon,\ S^*$
		1^{++}	A_1	Q_1	$D,\ E$
		2^{++}	A_2	K^*	$f,\ f'$

1 Background: The Three Key Ingredients

To understand G-parity, we need three concepts: **charge conjugation** (C), **isospin** (I), and **the isospin rotation operator** (R_2). Let's build them up.

1.1 Charge Conjugation (C-Parity)

The **charge conjugation operator** C transforms every particle into its antiparticle:

$$C \left| \text{particle} \right\rangle = \eta_C \left| \text{antiparticle} \right\rangle$$

It flips the sign of all additive quantum numbers (charge, baryon number, strangeness, etc.) but does **not** change mass, momentum, or spin.

Only particles that are their own antiparticles can be eigenstates of C. This means C-parity is defined only for neutral, flavourless states like:

$$\pi^0, \quad \rho^0, \quad \omega, \quad \eta, \quad \eta', \quad \phi, \quad \gamma$$

Charged particles like π^+ or ρ^+ are **not** eigenstates of C. This is the problem that G-parity solves.

For a $q\bar{q}$ meson (where the quark and antiquark are of the same flavour), C-parity is:

$$\boxed{C = (-1)^{L+S}}$$

where L is the orbital angular momentum and S is the total quark spin.

Origin: The charge conjugation operator swaps quark $\leftrightarrow$ antiquark. This is equivalent to exchanging the two particles, which produces:

- A factor $(-1)^L$ from the spatial wavefunction (exchanging positions in a state with angular momentum L)
- A factor $(-1)^{S+1}$ from the spin wavefunction (exchange of two spin-$\frac{1}{2}$ fermions, including Fermi sign)
- A factor (-1) from the intrinsic C-parity of a fermion–antifermion pair

These combine to give $C = (-1)^L \cdot (-1)^{S+1} \cdot (-1) = (-1)^{L+S}$.

1.2 Isospin (I)

Isospin is a quantum number that treats the u and d quarks as two states of the same particle, analogous to spin-$\frac{1}{2}$:

$$u = \left|\tfrac{1}{2},\ +\tfrac{1}{2}\right\rangle, \qquad d = \left|\tfrac{1}{2},\ -\tfrac{1}{2}\right\rangle$$

Mesons made from u and d quarks (and their antiquarks) form isospin multiplets:

- $I = 1$ (triplet): π^+, π^0, π^- or ρ^+, ρ^0, ρ^- etc.
- $I = 0$ (singlet): $\eta, \eta', \omega, \phi$ etc.

Isospin is conserved in the strong interaction.

1.3 The Problem With C Alone

Here's the issue: the π^+ decays strongly, and we'd like a quantum number like C that's conserved. But $C\,|\pi^+\rangle = |\pi^-\rangle$, which is a *different* particle. The π^+ is not an eigenstate

of C.

The solution is to combine C with an isospin rotation that maps $\pi^+ \leftrightarrow \pi^-$, so the *combined* operation brings the particle back to itself.

1.4 G-Parity: The Definition

The **G-parity operator** is defined as:

$$\boxed{G = C \cdot e^{i\pi I_2}}$$

where C is charge conjugation and $e^{i\pi I_2}$ is a rotation by π radians (180°) about the 2-axis in isospin space. We denote this rotation $R_2 \equiv e^{i\pi I_2}$.

The key idea: R_2 flips the sign of I_3 (swapping particle $\leftrightarrow$ antiparticle within a multiplet), while C handles the charge conjugation. Together, G can assign a definite quantum number to *every* member of an isospin multiplet (not just neutral ones).

The choice of the 2-axis is a convention. A rotation by π about I_2 sends $I_3 \to -I_3$, which maps:

$$\pi^+ \ (I_3 = +1) \ \longleftrightarrow \ \pi^- \ (I_3 = -1)$$

The combination $G = C \cdot R_2$ then maps $\pi^+ \to \pi^+$ (back to itself), making π^+ an eigenstate of G.

2 The Master Formula: $G = (-1)^{L+S+I}$

For a $q\bar{q}$ meson with definite L, S, and I, we can combine the C-parity result with the isospin rotation to obtain:

$$\begin{aligned} G &= (-1)^I \cdot C \qquad \text{(Eq. 4.60, proved in Part (b))} && (1) \\ &= (-1)^I \cdot (-1)^{L+S} \\ &= \boxed{(-1)^{L+S+I}} && (2) \end{aligned}$$

This is the formula we will use for all the mesons in Part (a). Note that $C = (-1)^{L+S}$ applies to the neutral member of the multiplet, but G applies to the **entire multiplet**.

3 Part (a): Computing G-Parity for Each Meson

For each meson, we read off L, S, and I from Table 4.6, compute $C = (-1)^{L+S}$, then $G = (-1)^I \cdot C$.

3.1 The Pion (π): $I = 1$, $\ell = 0$, $s = 0$, $J^{PC} = 0^{-+}$

From Table 4.6: $L = 0$, $S = 0$, $I = 1$.

Step 1: C-parity (from the neutral member π^0):

$$C = (-1)^{L+S} = (-1)^{0+0} = +1$$

This is consistent with $J^{PC} = 0^{-+}$ (the $+$ is the C-parity). ✓

Step 2: G-parity:

$$G = (-1)^I \cdot C = (-1)^1 \cdot (+1) = \boxed{-1}$$

Since G-parity is multiplicative and each pion has $G = -1$, a system of n pions has $G = (-1)^n$. This powerful selection rule tells us:

- Particles with $G = +1$ can only decay into an **even** number of pions
- Particles with $G = -1$ can only decay into an **odd** number of pions

3.2 The Rho (ρ): $I = 1$, $\ell = 0$, $s = 1$, $J^{PC} = 1^{--}$

From Table 4.6: $L = 0$, $S = 1$, $I = 1$.

C-parity:

$$C = (-1)^{L+S} = (-1)^{0+1} = -1$$

Consistent with $J^{PC} = 1^{--}$. ✓

G-parity:

$$G = (-1)^I \cdot C = (-1)^1 \cdot (-1) = \boxed{+1}$$

Consequence: Since $G(\rho) = +1$, the ρ can only decay into an **even** number of pions. Indeed, the dominant decay is $\rho \to \pi\pi$ (two pions). The decay $\rho \to 3\pi$ is forbidden by G-parity conservation.

3.3 The Omega (ω): $I = 0$, $\ell = 0$, $s = 1$, $J^{PC} = 1^{--}$

From Table 4.6: $L = 0$, $S = 1$, $I = 0$.

C-parity:

$$C = (-1)^{L+S} = (-1)^{0+1} = -1$$

Consistent with $J^{PC} = 1^{--}$. ✓

G-parity:

$$G = (-1)^I \cdot C = (-1)^0 \cdot (-1) = \boxed{-1}$$

Consequence: Since $G(\omega) = -1$, the ω can only decay into an **odd** number of pions. Indeed, the dominant decay is $\omega \to \pi^+\pi^-\pi^0$ (three pions). The decay $\omega \to 2\pi$ is forbidden by G-parity.

The ρ and ω have the same $J^{PC} = 1^{--}$ and nearly the same mass (≈ 770 vs 783 MeV). They differ only in isospin ($I = 1$ vs $I = 0$). This one difference in isospin flips the G-parity, completely changing the allowed decay modes:

Meson	I	G	Dominant decay
ρ	1	$+1$	$\rho \to 2\pi$
ω	0	-1	$\omega \to 3\pi$

G-parity is the reason these nearly identical particles have such different decay signatures.

3.4 The Eta (η): $I = 0$, $\ell = 0$, $s = 0$, $J^{PC} = 0^{-+}$

From Table 4.6: $L = 0$, $S = 0$, $I = 0$.

***C*-parity:**

$$C = (-1)^{L+S} = (-1)^{0+0} = +1$$

Consistent with $J^{PC} = 0^{-+}$. ✓

***G*-parity:**

$$G = (-1)^I \cdot C = (-1)^0 \cdot (+1) = \boxed{+1}$$

Consequence: $G(\eta) = +1$ means η decays into an even number of pions when decaying via the strong interaction. However, $\eta \to 2\pi$ is also forbidden by P and CP conservation (the η has $J^P = 0^-$, same as a single pion, and a two-pion state with $J = 0$ has $P = +1$). The main hadronic decay is $\eta \to 3\pi$, which violates isospin (the strong interaction doesn't quite conserve isospin perfectly due to u-d mass differences).

3.5 The Eta Prime (η'): $I = 0$, $\ell = 0$, $s = 0$, $J^{PC} = 0^{-+}$

The η' has the same quantum numbers as η (just a different flavour mixing). From Table 4.6: $L = 0$, $S = 0$, $I = 0$.

***C*-parity:**

$$C = (-1)^{L+S} = (-1)^0 = +1$$

***G*-parity:**

$$G = (-1)^I \cdot C = (-1)^0 \cdot (+1) = \boxed{+1}$$

3.6 The Phi (ϕ): $I = 0$, $\ell = 0$, $s = 1$, $J^{PC} = 1^{--}$

From Table 4.6: $L = 0$, $S = 1$, $I = 0$.

***C*-parity:**

$$C = (-1)^{L+S} = (-1)^{0+1} = -1$$

Consistent with $J^{PC} = 1^{--}$. ✓

***G*-parity:**

$$G = (-1)^I \cdot C = (-1)^0 \cdot (-1) = \boxed{-1}$$

Consequence: Like the ω, the ϕ has $G = -1$, so it should decay to an odd number of pions. However, the ϕ is predominantly $s\bar{s}$, so the OZI rule suppresses its decay to pions. Instead, $\phi \to K\bar{K}$ dominates (this is not a pionic decay, so G-parity doesn't directly constrain it). The subdominant decay $\phi \to 3\pi$ is consistent with $G = -1$.

3.7 The f_2 Meson: $I = 0$, $\ell = 1$, $s = 1$, $J^{PC} = 2^{++}$

In Table 4.6, the f (or f_2 in modern notation, since $J = 2$) appears in the $\ell = 1$, $s = 1$ row with $I = 0$.
From Table 4.6: $L = 1$, $S = 1$, $I = 0$.

C-parity:

$$C = (-1)^{L+S} = (-1)^{1+1} = +1$$

Consistent with $J^{PC} = 2^{++}$. ✓

G-parity:

$$G = (-1)^I \cdot C = (-1)^0 \cdot (+1) = \boxed{+1}$$

Consequence: $G(f_2) = +1$ means f_2 decays into an even number of pions. Indeed, the dominant decay mode is $f_2 \to \pi\pi$ (two pions).

3.8 Summary Table for Part (a)

Meson	I	L	S	J^{PC}	$C = (-1)^{L+S}$	$G = (-1)^I C$
π	1	0	0	0^{-+}	$+1$	$\mathbf{-1}$
ρ	1	0	1	1^{--}	-1	$\mathbf{+1}$
ω	0	0	1	1^{--}	-1	$\mathbf{-1}$
η	0	0	0	0^{-+}	$+1$	$\mathbf{+1}$
η'	0	0	0	0^{-+}	$+1$	$\mathbf{+1}$
ϕ	0	0	1	1^{--}	-1	$\mathbf{-1}$
f_2	0	1	1	2^{++}	$+1$	$\mathbf{+1}$

4 Part (b): Proving $R_2 \left| I\ 0 \right\rangle = (-1)^I \left| I\ 0 \right\rangle$

This is the mathematical heart of the problem. We need to show that the isospin rotation operator $R_2 = e^{i\pi I_2}$, when applied to the state $|I,\ I_3 = 0\rangle$, gives $(-1)^I$ times that same state.

4.1 The Strategy

The operator $R_2 = e^{i\pi I_2}$ is a rotation by π about the 2-axis in isospin space. Isospin obeys the *exact same* mathematics as angular momentum (the SU(2) algebra). So we can use everything we know about angular momentum rotations.

The isospin operators I_1, I_2, I_3 satisfy:

$$[I_i, I_j] = i\epsilon_{ijk} I_k$$

These are identical to the commutation relations of angular momentum J_1, J_2, J_3. All results about rotation operators $e^{i\theta J_n}$ acting on $|j, m\rangle$ states carry over directly to $e^{i\theta I_n}$ acting on $|I, I_3\rangle$ states.

4.2 What Does R_2 Do to $|I, I_3\rangle$?

A rotation by π about the 2-axis (the y-axis, in the angular momentum analogy) has a well-known effect. Using the Wigner d-matrix:

For a state $|j, m\rangle$, a rotation by angle β about the y-axis gives:

$$e^{-i\beta J_y} |j, m\rangle = \sum_{m'} d^{(j)}_{m'm}(\beta) \; |j, m'\rangle$$

where $d^{(j)}_{m'm}(\beta)$ are the Wigner (small) d-matrix elements.

For $\beta = \pi$ (a 180° rotation), there is a known identity:

$$d^{(j)}_{m'm}(\pi) = (-1)^{j-m} \, \delta_{m',-m}$$

This means a π-rotation about the y-axis maps $|j, m\rangle \to (-1)^{j-m} |j, -m\rangle$.

Note that in Griffiths' convention, $R_2 = e^{i\pi I_2}$, which corresponds to $e^{-i\pi J_y}$ with the identification $I_2 \leftrightarrow -J_y$ (or equivalently, the rotation is in the *opposite* sense). The standard QM rotation operator is $e^{-i\beta J_y}$. In either convention, the final result for $I_3 = 0$ states is the same, because the sign only affects the phase for $m \neq 0$ states. For $m = 0$, we always get $(-1)^j$.

4.3 Applying to the $I_3 = 0$ State

Now we set $m = 0$ (i.e., $I_3 = 0$) in the rotation formula:

Using the d-matrix identity for $\beta = \pi$:

$$R_2 |I, 0\rangle = e^{i\pi I_2} |I, 0\rangle = (-1)^{I-0} |I, -0\rangle = (-1)^I |I, 0\rangle$$

Since $-0 = 0$, the state maps back to itself with a factor of $(-1)^I$.

$$\boxed{R_2 \left|I,\ 0\right\rangle = (-1)^I \left|I,\ 0\right\rangle}$$

In words: The isospin rotation by π about the 2-axis leaves the $I_3 = 0$ state as an eigenstate, with eigenvalue $(-1)^I$. For $I = 0$ states, R_2 gives $+1$. For $I = 1$ states, R_2 gives -1.

4.4 Verifying for Small Cases

Let's verify this explicitly for the simplest cases.

4.4.1 $I = 0$ (Singlet)

The state $|0, 0\rangle$ is a scalar under isospin rotations. Any rotation leaves it unchanged:

$$R_2 \left|0, 0\right\rangle = (+1) \left|0, 0\right\rangle = (-1)^0 \left|0, 0\right\rangle \quad \checkmark$$

4.4.2 $I = 1$ (Triplet, e.g., Pions)

For $I = 1$, isospin behaves exactly like spin-1. The I_2 operator in the $|1, m\rangle$ basis (with $m = +1, 0, -1$) is represented by:

$$I_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}$$

We need $e^{i\pi I_2}$. The eigenvalues of I_2 for $I = 1$ are $+1, 0, -1$. One can compute the matrix exponential (or use the d-matrix formula directly). The result for the $|1, 0\rangle$ state is:

$$R_2 \left|1, 0\right\rangle = (-1)^1 \left|1, 0\right\rangle = - \left|1, 0\right\rangle \quad \checkmark$$

This means the π^0 (which is the $I_3 = 0$ member of the pion triplet) picks up a factor of -1 under R_2.

4.5 Justifying Equation 4.60: $G = (-1)^I C$

The G-parity operator is:

$$G = C \cdot R_2 = C \cdot e^{i\pi I_2}$$

We apply G to the *neutral member* of an isospin multiplet, $|I, 0\rangle$, since this is the member that is an eigenstate of C.

Step 1: Apply R_2 first:

$$R_2 \left|I, 0\right\rangle = (-1)^I \left|I, 0\right\rangle \qquad \text{(just proved above)}$$

Step 2: Now apply C:

$$G \left|I, 0\right\rangle = C \cdot R_2 \left|I, 0\right\rangle = C \cdot (-1)^I \left|I, 0\right\rangle = (-1)^I \cdot C \left|I, 0\right\rangle = (-1)^I \cdot \eta_C \left|I, 0\right\rangle$$

where $\eta_C = C$ is the charge conjugation eigenvalue of the neutral member.

Step 3: Reading off the G-parity eigenvalue:

$$G\,|I,0\rangle = \underbrace{(-1)^I \cdot C}_{\text{G-parity eigenvalue}}\ |I,0\rangle$$

Therefore:

$$\boxed{G = (-1)^I\, C}$$

This is Equation 4.60. The crucial point is that this value of G is the same for **every** member of the isospin multiplet (not just the neutral one), because G commutes with the strong interaction Hamiltonian and all members of a multiplet have the same G-parity.

We chose to evaluate G on $|I,0\rangle$ because:

1. Only the neutral member has a well-defined C-parity eigenvalue
2. We proved that $R_2\,|I,0\rangle = (-1)^I\,|I,0\rangle$, so R_2 simply contributes a phase
3. Since G is the same for all multiplet members (by isospin symmetry), evaluating it on the neutral member determines G for the entire multiplet

5 Summary of Key Results

Part (a): The G-parities of the mesons, computed using $G = (-1)^{L+S+I}$:

Meson	G	Decay constraint	Observed decay
π	-1	—	(stable under strong)
ρ	$+1$	even # of π's	$\rho \to 2\pi$ ✓
ω	-1	odd # of π's	$\omega \to 3\pi$ ✓
η	$+1$	even # of π's	(see note)
η'	$+1$	even # of π's	$\eta' \to \eta\pi\pi$
ϕ	-1	odd # of π's	$\phi \to 3\pi$ ✓
f_2	$+1$	even # of π's	$f_2 \to 2\pi$ ✓

Part (b): We proved that $R_2\,|I,0\rangle = (-1)^I\,|I,0\rangle$ using the Wigner d-matrix identity for 180° rotations. This directly implies:

$$G = (-1)^I \cdot C$$

because applying $G = C \cdot R_2$ to the neutral member of a multiplet gives a factor $(-1)^I$ from R_2 and a factor C from charge conjugation, and this G-parity eigenvalue is the same for all members of the multiplet.

6 Appendix: Quick Reference Formulae

For a $q\bar{q}$ meson with orbital angular momentum L, total quark spin S, total angular momentum J, and isospin I:

$$J = |L - S|, \ldots, L + S$$

$$P = (-1)^{L+1} \qquad \text{(Parity)}$$

$$C = (-1)^{L+S} \qquad (C\text{-parity, neutral members only})$$

$$G = (-1)^{L+S+I} = (-1)^I \cdot C \qquad (G\text{-parity, entire multiplet})$$

G-parity selection rule for strong decays to pions:

$$G_{\text{parent}} = (-1)^n \qquad \text{where } n = \text{number of pions in the final state}$$

(Each pion has $G = -1$, and G is multiplicative.)